JOSUÉ JIMÉNEZ VILLARREAL

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Aeronautical Engineer with a mechatronics foundation. Skilled in analytical and simulation-driven design, along with the practical side of manufacturing, including prototyping, GD&T documentation, and design-for-manufacturability. Comfortable validating a concept in simulation and then refining it with cross-functional teams.

PROFESSIONAL EXPERIENCE

Lean Challenge Intern — ABB

June 2026 – August 2026

- Part of the winning team in ABB's international Lean Challenge competition.
- Redesigned production components and optimized assembly processes, reducing cycle time by up to 70% and improving ergonomics through cross-functional collaboration.
- Utilized DMAIC methodology, CAD modeling, and validation techniques to drive continuous improvement initiatives.

Mechanical Design Intern — John Deere

August 2024 – May 2026

- Designed and modified sheet metal and plate brackets, castings, and injection-molded parts for agricultural machinery, balancing load, weight, material, and cost constraints.
- Created 3D models (Creo parametric) and drawings with GD&T, ensuring compliance with engineering standards.
- Performed structural validation using FEA (SimSolid), iterating designs based on analysis results and design reviews.
- Participated in design reviews and redesigns based on field test failures to improve reliability.
- Collaborated with external suppliers (plastic injection) to refine designs for manufacturability.
- Assisted in managing product structures and supporting BOM definition using PLM (PTC Windchill).

EDUCATION

B.S. in Aeronautical Engineering — Universidad Autónoma de Nuevo León

August 2020 – June 2026

- Selected among the top 100 students with the highest academic performance within the Faculty of Mechanical and Electrical Engineering (FIME) during the August–December 2025 semester.

Technical Degree in Mechatronics — Universidad Autónoma de Nuevo León

July 2021 – December 2022

SKILLS

- Languages: English (Fluent)
- CAD: Creo Parametric, SolidWorks, AutoCAD
- CAE: SimSolid, ANSYS (Structural, Fluent, ACP)
- Programming & Analysis: MATLAB, Python
- PLM: PTC Windchill

COURSES & CERTIFICATIONS

- Certified SolidWorks Associate (CSWA)
- Lean Manufacturing — Udemy
- Failure Analysis — UANL

ENGINEERING PROJECTS

- Designed aircraft structures using analytical calculations, CFD, wind tunnel testing, and FEA to optimize aerodynamic and structural performance while reducing weight and material usage by 25%.
- Designed and fabricated composite laminates using carbon fiber, fiberglass, and honeycomb cores, optimizing stacking sequence, fiber orientation, and thickness through ANSYS ACP simulations, then cured parts via vacuum bagging, autoclave, and manual layup techniques.
- Created and validated prototypes, fixtures, and enclosures using Creo, SolidWorks, and FDM 3D printing to verify fit, assembly, tolerances, and functionality.
- Characterized magnetorheological elastomers through vibration testing and MATLAB/Python data analysis to evaluate material behavior under magnetic activation.